



Negative Sampling and Evaluation in Dynamic Graph Learning

From pairwise graphs to dynamic hypergraphs

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☐ Overview of Dynamic Graph Learning

☐ Paper - Towards Better Evaluation for Dynamic Link Prediction

- Understanding Dynamic Graph Datasets
- A New Baseline for Dynamic Link Prediction
- Various Negative Sampling Strategies

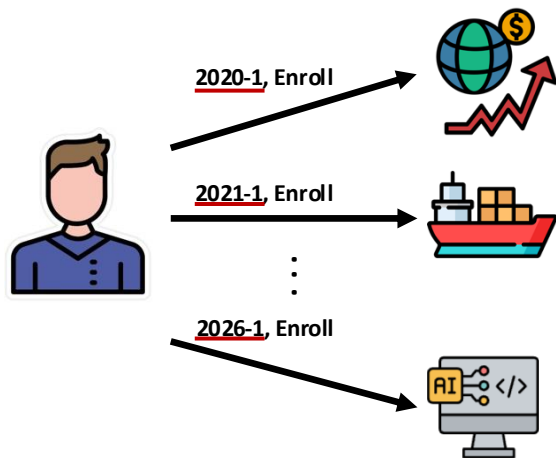
☐ Research - Negative Sampling in Dynamic Hypergraphs

- Overview of Dynamic Hypergraph Learning
- Observed Challenges and Negative Sampling Strategies
- Planned Experiments

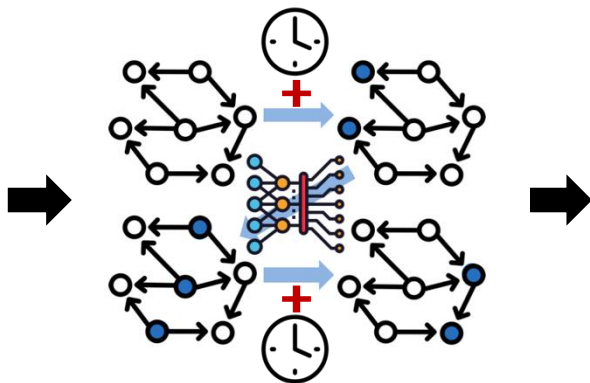
Overview of Dynamic Graph Learning

- Unlike static graphs, each interaction is associated with a timestamp
- Models both structural information and temporal patterns
- Learns node embeddings that evolve over time

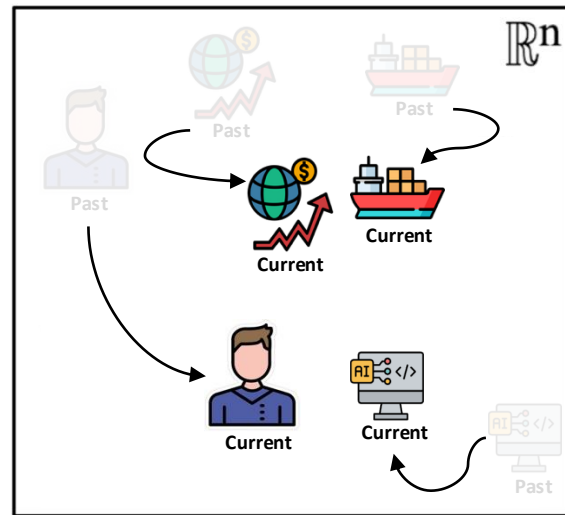
Course Enrollment Dynamic Graph



Dynamic Graph Learning Models



Time-Evolving Node Embeddings



Two Types of Dynamic Graph Learning

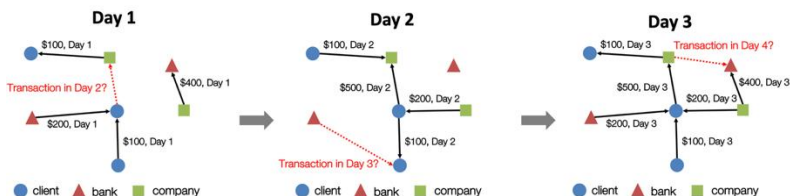
□ Discrete-time dynamic graphs

- A sequence of snapshots aggregated over fixed time gaps
- Each snapshot as an individual static graph, $G_T = (G_{t_1}, G_{t_2}, \dots, G_{t_n})$

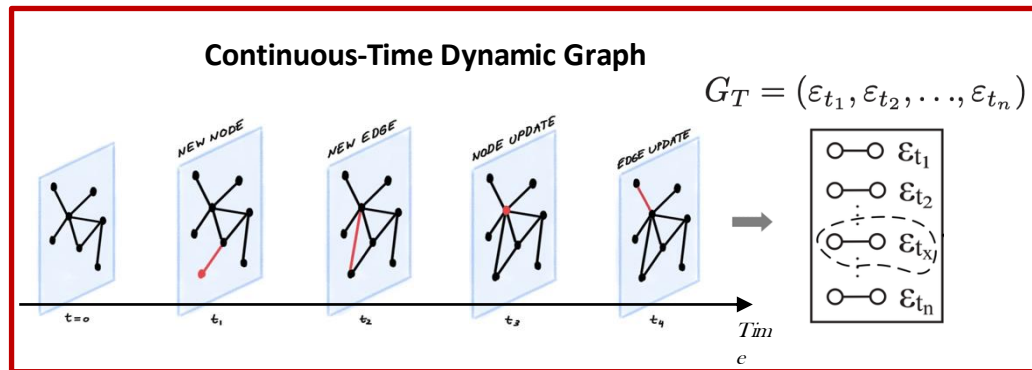
□ Continuous-time dynamic graphs

- Event-based graph representation
- Chronological ordering of graph edges as time-stamped events, $\varepsilon_t = (i, j, t)$

Discrete-Time Dynamic Graph



Continuous-Time Dynamic Graph



Two Downstream Tasks of Dynamic Graph Learning

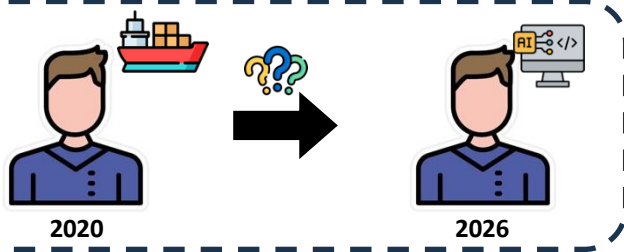
☐ Dynamic node classification

- Prediction of changes in the labels of nodes at a given time

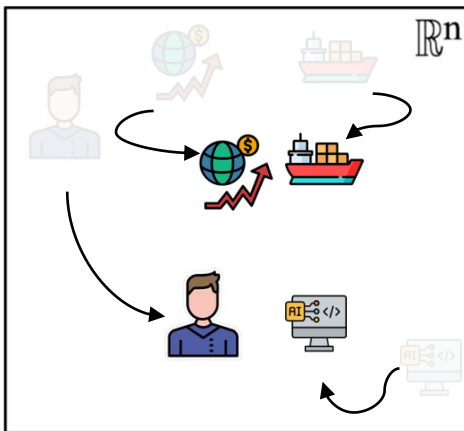
☐ Future Link prediction

- Prediction of whether nodes u and v are connected (interact) at a specific time

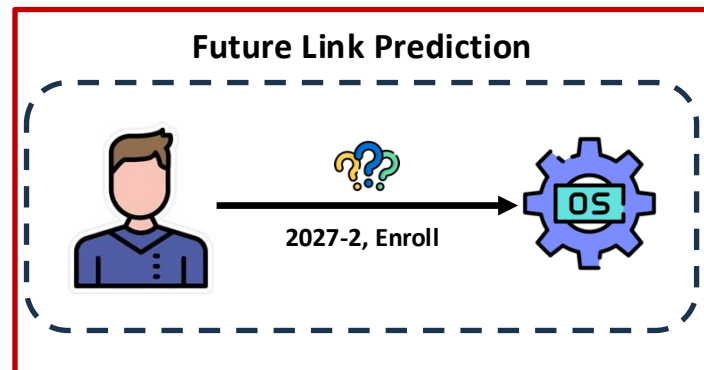
Dynamic Node Classification



Learned Node Embeddings



Future Link Prediction



Training Process in Dynamic Graph Learning

- Trained with negative sampling, as in static link prediction
 - Generating a negative sample by randomly replacing one node
- Binary classification using BCE loss, $\mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{neg}}$
 - Positive edge $\rightarrow$ prediction probability close to 1
 - Negative edge $\rightarrow$ prediction probability close to 0

Direct Message History



Training Pair

Positive Sample (Observed)



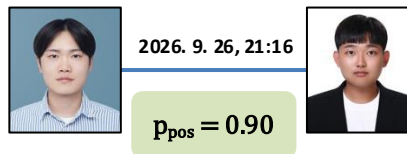
Replaced with
a randomly sampled node

Negative Sample

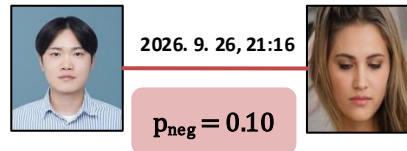


Prediction Scores

Positive Test Edge



Negative Sample



Training Objective

BCE Loss

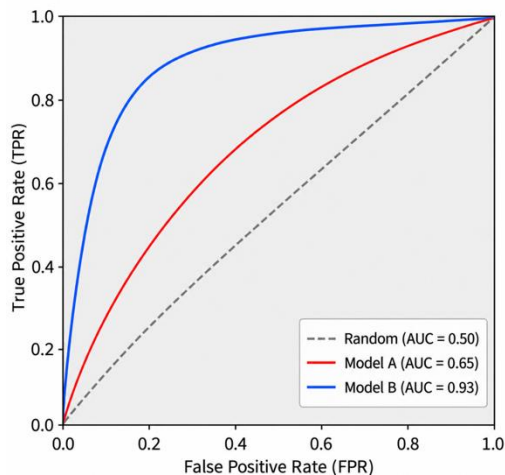
$$\mathcal{L} = \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{neg}} \\ = -\log p_{\text{pos}} - \log(1 - p_{\text{neg}})$$

Update
Model Parameters

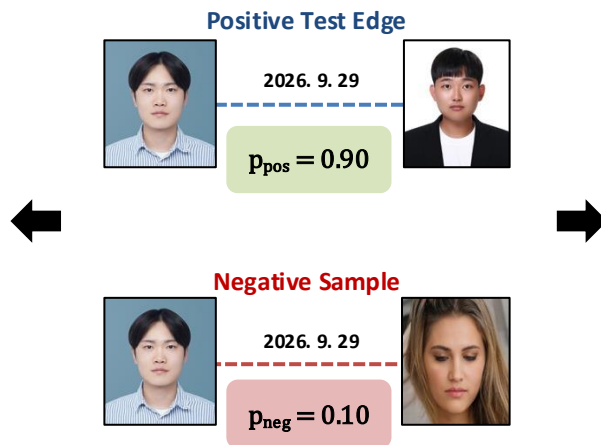
Evaluation Process in Dynamic Graph Learning

- Evaluation with positive and sampled negative edges
- Performance measurement using AUC and AP
 - AUC: Probability that a positive edge receives a higher score than a negative edge
 - AP: Average precision at ranks where positive edges are retrieved

Area Under the ROC Curve



Example Prediction Scores



Average Precision

Rank	Score	Label	Precision@k
1	0.95	P	$1/1 = 1.00$
2	0.90	N	$1/2 = 0.50$
3	0.80	P	$2/3 = 0.67$
4	0.60	P	$3/4 = 0.75$
5	0.20	N	$3/5 = 0.60$

$$AP = \frac{1.00 + 0.67 + 0.75}{3} \approx 0.81$$

Near-perfect SOTA performance

- Near-perfect SOTA performance on most existing benchmark datasets

Task		Methods	Reddit	Wikipedia	MOOC	Social Evo.	Enron	UCI
AUC	Transductive	DynAERNN	83.37 $\pm$ 1.48	71.00 $\pm$ 1.10	89.34 $\pm$ 0.24	67.78 $\pm$ 0.80	63.11 $\pm$ 1.13	83.72 $\pm$ 1.79
		JODIE	87.71 $\pm$ 0.02	88.43 $\pm$ 0.02	90.50 $\pm$ 0.01 [†]	89.78 $\pm$ 0.04	89.36 $\pm$ 0.06	74.63 $\pm$ 0.11
		DyRep	67.36 $\pm$ 1.23	77.40 $\pm$ 0.13	90.49 $\pm$ 0.03	90.85 $\pm$ 0.01 [†]	96.71 $\pm$ 0.04 [†]	95.23 $\pm$ 0.25 [†]
		VGRNN	51.89 $\pm$ 0.92	71.20 $\pm$ 0.65	90.03 $\pm$ 0.32	78.28 $\pm$ 0.69	93.84 $\pm$ 0.58	89.43 $\pm$ 0.27
		EvolveGCN	58.42 $\pm$ 0.52	60.48 $\pm$ 0.47	50.36 $\pm$ 0.85	60.36 $\pm$ 0.65	74.02 $\pm$ 0.31	78.30 $\pm$ 0.22
		TGAT	96.65 $\pm$ 0.06 [†]	96.36 $\pm$ 0.05 [†]	72.09 $\pm$ 0.29	56.63 $\pm$ 0.55	60.88 $\pm$ 0.37	77.67 $\pm$ 0.27
		CAW-N-mean	99.97 $\pm$ 0.01*	99.91 $\pm$ 0.04*	91.99 $\pm$ 0.72*	94.12 $\pm$ 0.15*	93.53 $\pm$ 0.73	95.90 $\pm$ 0.71
		CAW-N-attn	99.98 $\pm$ 0.01*	99.89 $\pm$ 0.03*	92.38 $\pm$ 0.58*	94.79 $\pm$ 0.16*	95.93 $\pm$ 0.39	98.45 $\pm$ 0.49*
AP	Transductive	DynAERNN	85.58 $\pm$ 2.12	76.58 $\pm$ 1.41	89.29 $\pm$ 0.49	66.58 $\pm$ 1.84	60.90 $\pm$ 2.70	84.95 $\pm$ 2.13
		JODIE	91.14 $\pm$ 0.01	91.39 $\pm$ 0.04	91.19 $\pm$ 0.03 [†]	89.22 $\pm$ 0.01	91.94 $\pm$ 0.01	80.27 $\pm$ 0.08
		DyRep	67.54 $\pm$ 2.02	77.36 $\pm$ 0.25	90.49 $\pm$ 0.03	94.48 $\pm$ 0.01 [†]	97.14 $\pm$ 0.07 [†]	95.29 $\pm$ 0.13 [†]
		VGRNN	50.87 $\pm$ 0.81	67.66 $\pm$ 0.89	83.70 $\pm$ 0.56	78.66 $\pm$ 0.67	94.02 $\pm$ 0.52	82.23 $\pm$ 0.56
		EvolveGCN	54.49 $\pm$ 0.73	55.84 $\pm$ 0.37	51.80 $\pm$ 0.46	56.90 $\pm$ 0.54	69.72 $\pm$ 0.49	81.63 $\pm$ 0.23
		TGAT	98.38 $\pm$ 0.01 [†]	96.65 $\pm$ 0.06 [†]	69.75 $\pm$ 0.23	57.37 $\pm$ 1.18	57.37 $\pm$ 0.18	60.25 $\pm$ 0.31
		CAW-N-mean	99.96 $\pm$ 0.01*	99.91 $\pm$ 0.03*	92.05 $\pm$ 0.88	95.90 $\pm$ 0.09*	94.93 $\pm$ 0.39	95.89 $\pm$ 0.87
		CAW-N-attn	99.99 $\pm$ 0.01*	99.89 $\pm$ 0.03*	92.23 $\pm$ 0.76*	96.37 $\pm$ 0.09*	96.13 $\pm$ 0.37	98.86 $\pm$ 0.38*



Towards Better Evaluation for Dynamic Link Prediction

Farimah Poursafaei, Shenyang Huang, Kellin Pelrine, Reihaneh Rabbany

McGill University School of Computer Science, Mila – Quebec AI Institute

NeurIPS '22, Datasets and Benchmarks Track

☐ Overview of Dynamic Graph Learning

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- Understanding Dynamic Graph Datasets
- A New Baseline for Dynamic Link Prediction
- Revisiting Negative Sampling in Dynamic Graphs

☐ Research - Negative Sampling in Dynamic Hypergraphs

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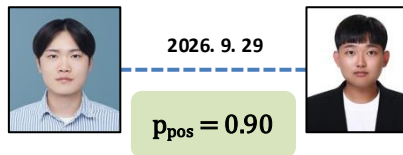
Two Questions Behind Near-Perfect Performance

- ☐ Are existing benchmark datasets too simplistic?
- ☐ Is the current evaluation sufficient to differentiate methods?
 - Is the current evaluation challenging enough?

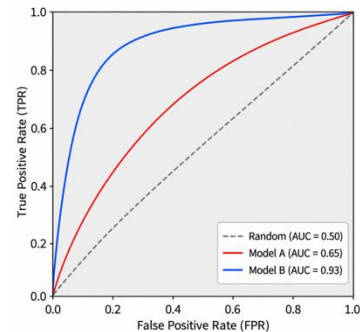
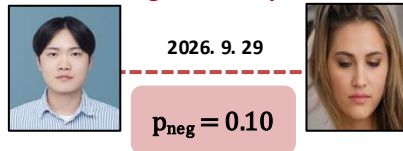
Too Simple?



Positive Sample (Observed)



Negative Sample

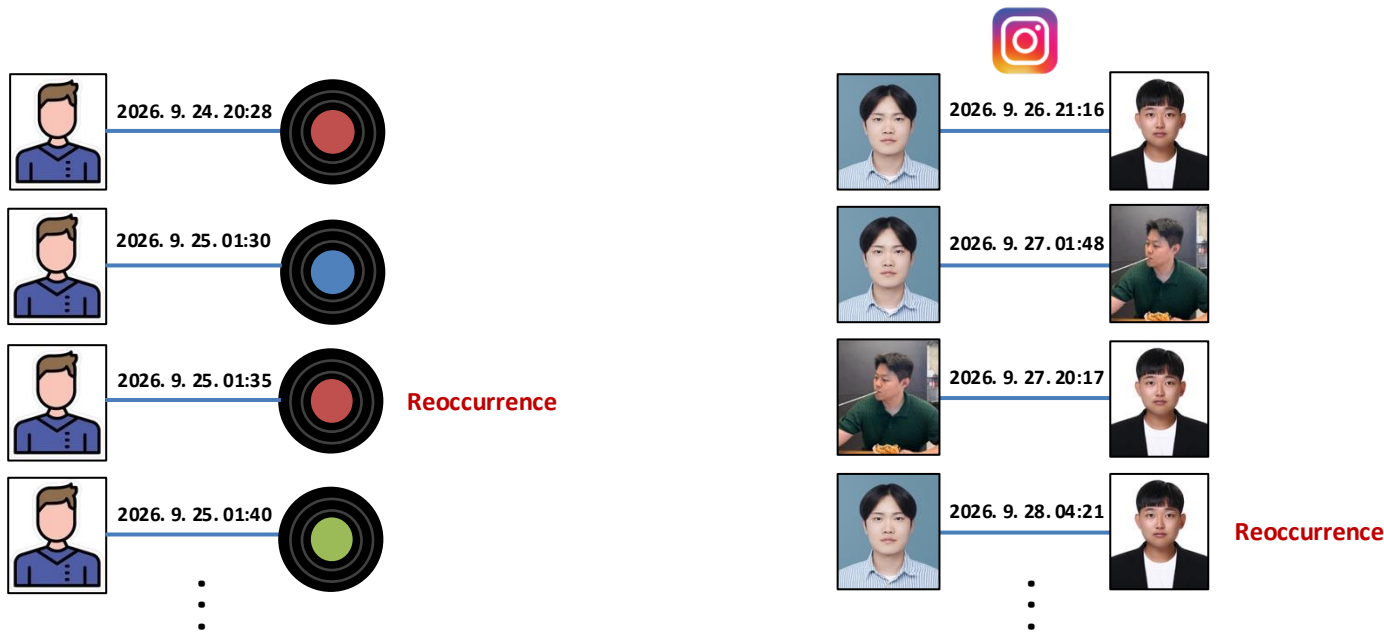


Rank	Score	La bel	Pre cision@k
1	0.95	P	1/1 = 1.00
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4	0.60	P	3/4 = 0.75
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$$AP = \frac{1.00 + 0.67 + 0.75}{3} \approx 0.81$$

Understanding Dynamic Graph Datasets

- Continuous evolution of relationships over time
 - Diverse temporal evolution patterns across datasets
- **Frequent edge reoccurrence with varying proportions across datasets**



Temporal Edge Appearance (TEA)

- Visualization of repeated and newly observed edges at each timestamp

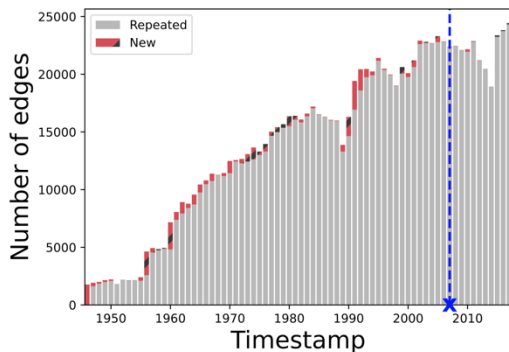
- Repeated: edges observed at previous timestamps

- New: edges observed for the first time

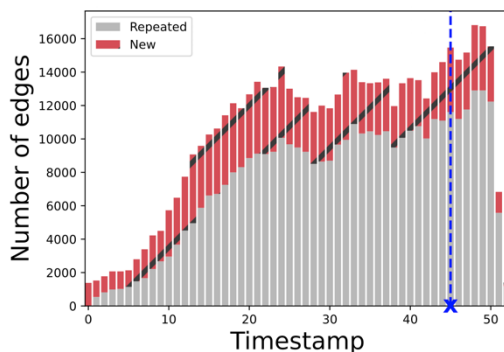
- Novelty index for quantifying newly observed edges

- Average proportion of new edges across timestamps, $\frac{1}{T} \sum_{t=1}^T \frac{|E^t \setminus E_{seen}^t|}{|E^t|}$

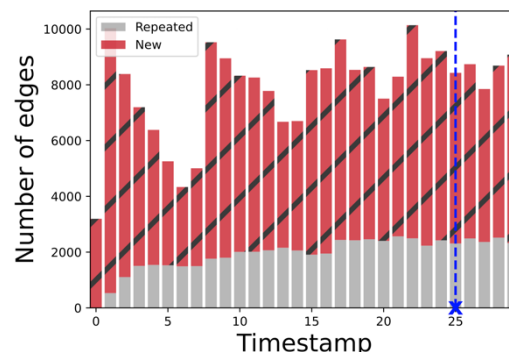
Dataset (Novelty Index)



UN Vote (0.03)



LastFM (0.28)



MOOC (0.75)

Temporal Edge Traffic (TET)

- Visualization of edge reoccurrence patterns across the train–test split

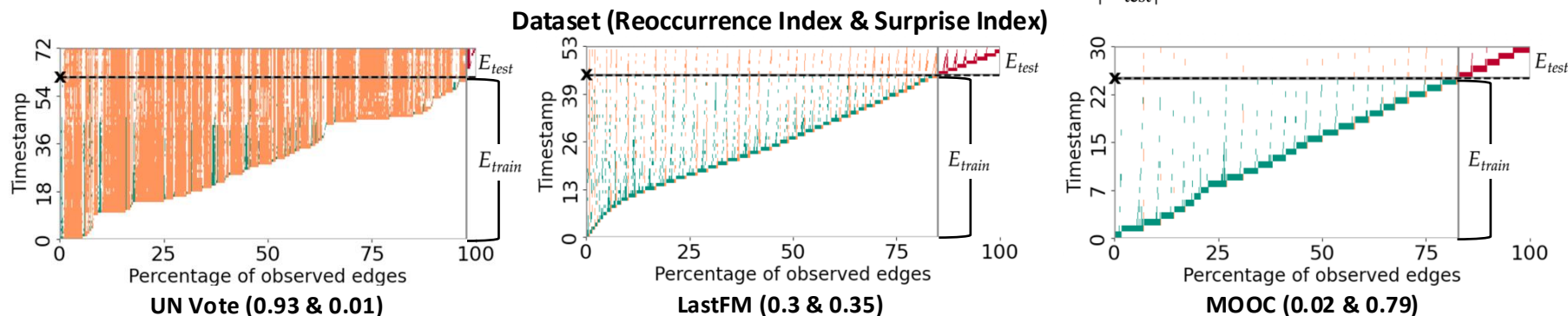
- Training-only edges (green) — $E_{train} \setminus E_{test}$

- Edges appearing in both training and test (orange) — $E_{train} \cap E_{test}$

- Test-only unseen edges (red) — $E_{test} \setminus E_{train}$

- Reoccurrence index for quantifying training edges reappearing in test, $\frac{|E_{train} \cap E_{test}|}{|E_{train}|}$

- Surprise index for quantifying test edges unseen during training, $\frac{|E_{test} \setminus E_{train}|}{|E_{test}|}$



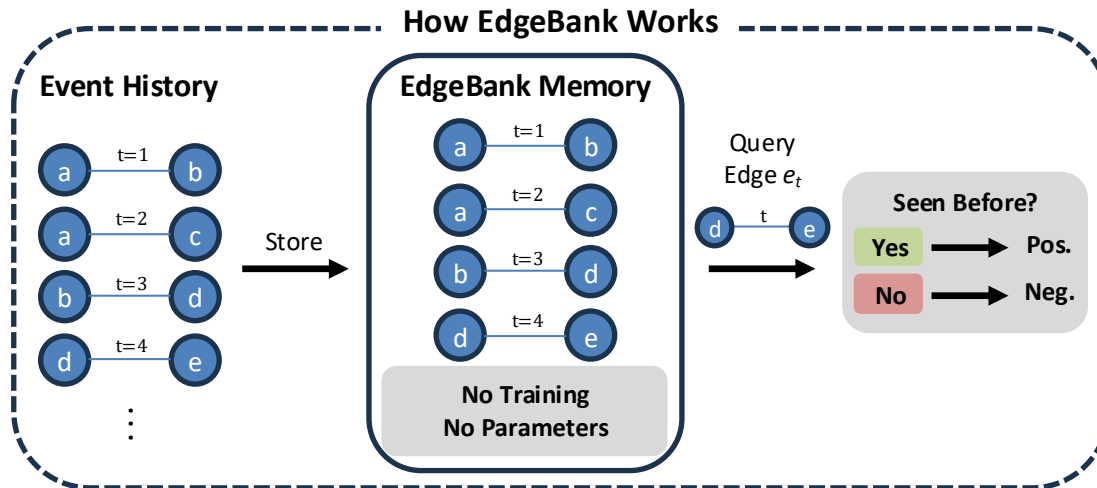
High reoccurrence + low surprise → Many test edges have already been observed during training

A New Baseline – EdgeBank (cont.)

□ A parameter-free baseline based purely on edge memorization

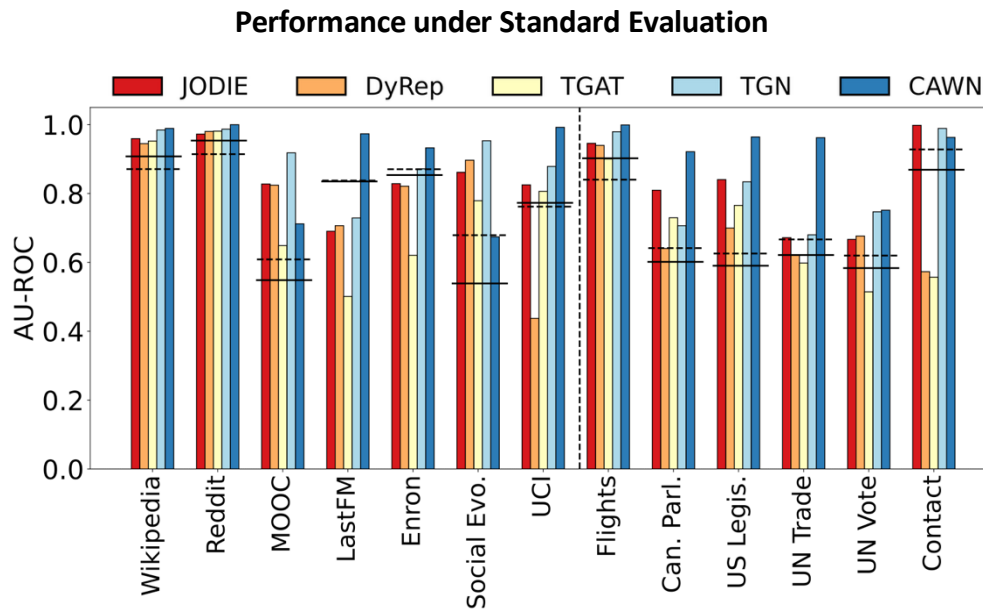
■ Seen before → Positive

■ Never seen → Negative



A New Baseline – EdgeBank

- Two memory strategies
 - ∞ : all previously observed edges, tw: edges observed within a recent time window
- Some benchmark datasets are too simple



Limitations of Evaluation with Random Negatives

- High train-test edge reoccurrence in some dynamic graphs
- Previously observed edges are rarely sampled as negatives in sparse dynamic graphs
- Potentially easy discrimination between seen positives and unseen negatives



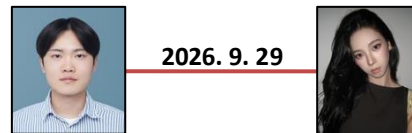
Training Phase

Input for Edge Prediction



VS

Negative Samples



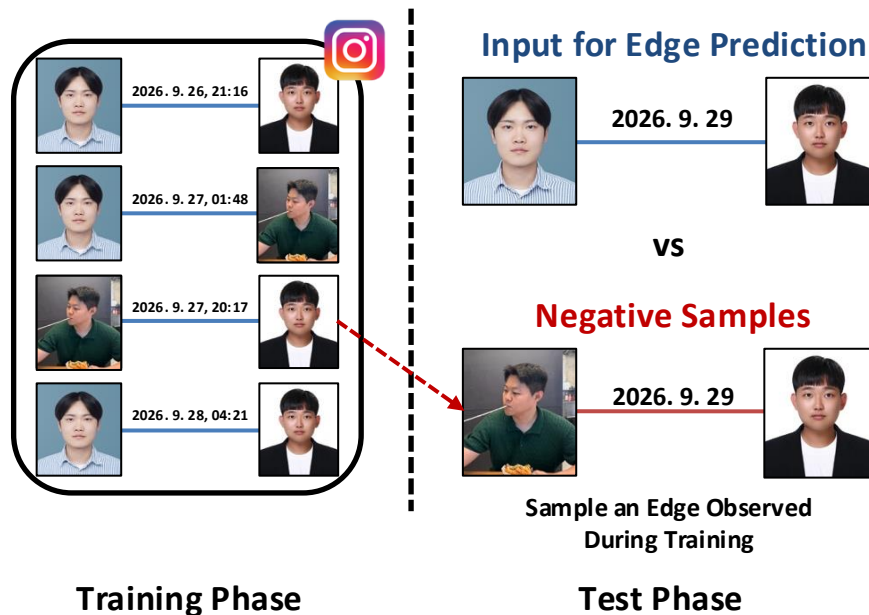
Never Seen Before

Replaced with
a randomly sampled node

Test Phase

Revisiting Negative Sampling in Dynamic Graphs (cont.)

- Historical NS → Previously observed edge, but absent at the current timestamp



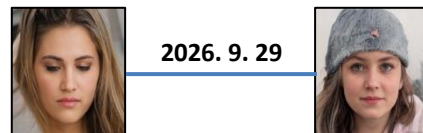
Revisiting Negative Sampling in Dynamic Graphs

- Inductive NS → Edges unseen during training, but observed earlier in the test phase



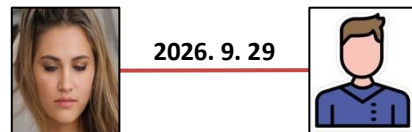
Training Phase

Input for Edge Prediction

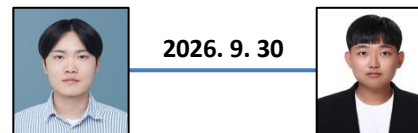


VS

Negative Samples

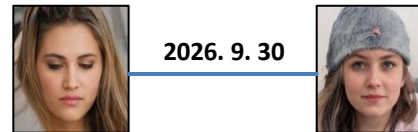


Input for Edge Prediction



VS

Negative Samples



Sample an Edge Unseen in Training
but Observed Earlier in Test

Test Phase

Experiment Settings

☐ 5 SOTA dynamic graph models

- JODIE (KDD 2019)
- DyRep (ICLR 2019)
- TGAT (ICLR 2020)
- TGN (ICML 2020)
- CAWN (ICLR 2021)

☐ 13 dynamic graph datasets

- 7 widely used benchmark datasets, 6 newly introduced datasets from diverse domains

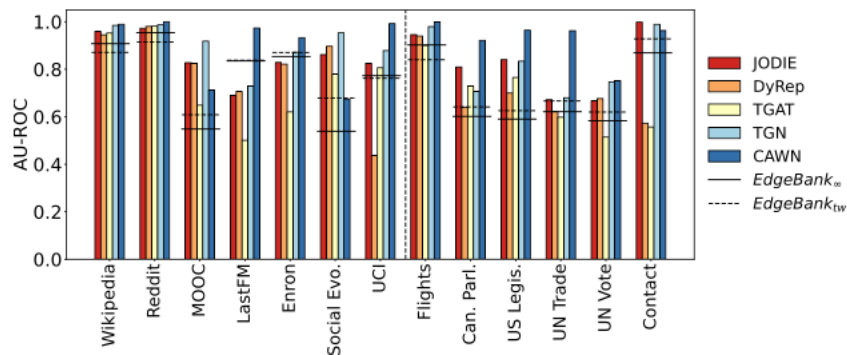
☐ 70% / 15% / 15% chronological train–validation–test split

☐ 3 negative sampling settings

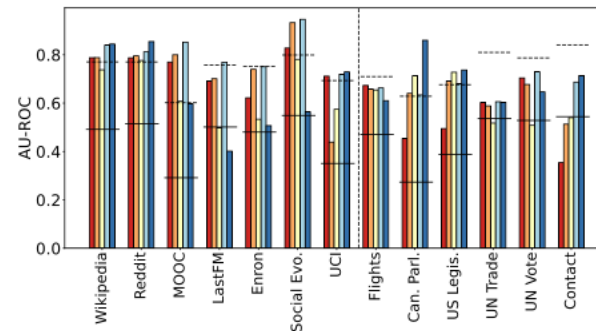
- Random / Historical / Inductive

Performance under Historical NS

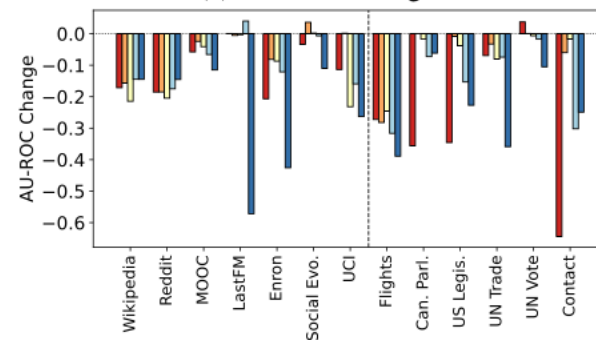
- Performance degradation compared with random NS
- Strong competitiveness of EdgeBank_{tw}



(a) Standard random setting.



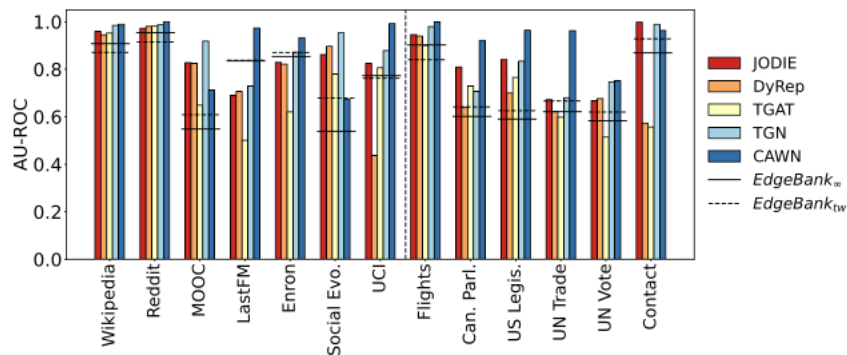
(b) Historical setting.



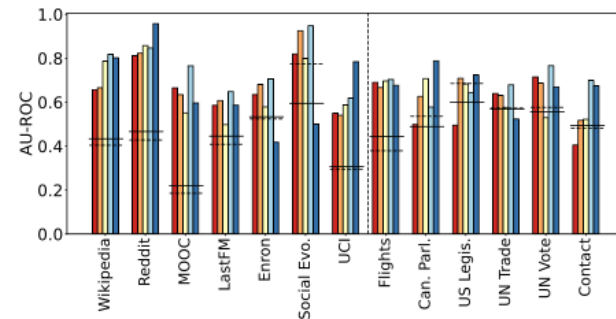
(d) Historical compared to random NS.

Performance under Inductive NS

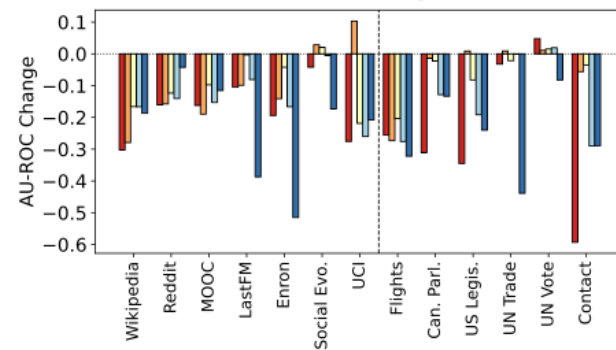
- Performance degradation compared with random NS
- Greater challenge for unseen-edge generalization



(a) Standard random setting.



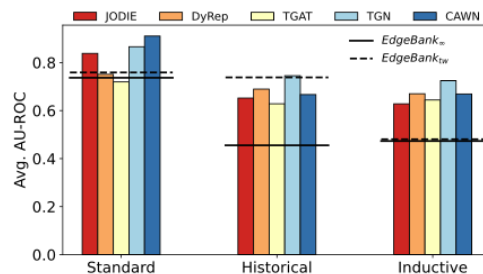
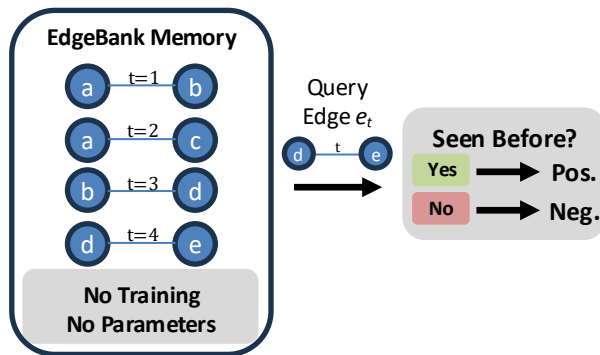
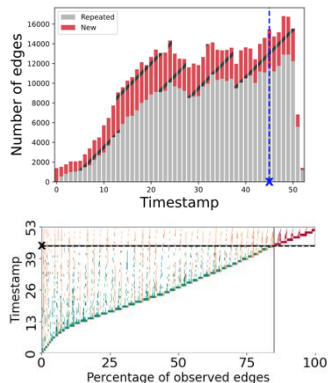
(c) Inductive setting.



(e) Inductive compared to random NS.

Conclusion

- Two key questions behind near-perfect performance
 - Simplicity of existing benchmark datasets
 - Sufficiency of current evaluation for differentiating methods
- Temporal pattern analysis with TEA and TET
- EdgeBank as a simple memorization-based baseline
- More challenging evaluation with historical and inductive negative sampling



Negative Sampling in Dynamic Hypergraph Learning

How can better negative sampling improve dynamic hyperedge prediction?

☐ Overview of Dynamic Graph Learning

☐ Paper - Towards Better Evaluation for Dynamic Link Prediction

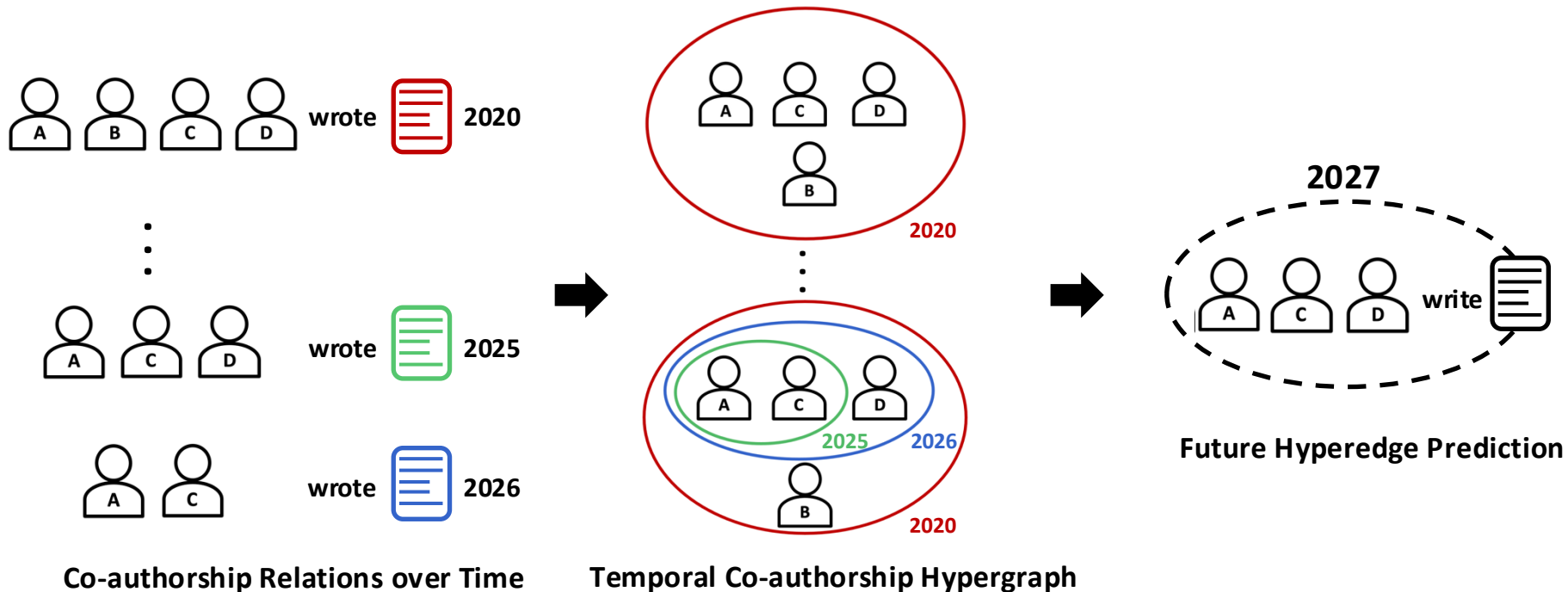
- ☐ Understanding Dynamic Graph Datasets
- ☐ A New Baseline for Dynamic Link Prediction
- ☐ Revisiting Negative Sampling in Dynamic Graphs

☐ **Research - Negative Sampling in Dynamic Hypergraphs**

- ☒ Overview of Dynamic Hypergraph Learning
- ☒ Observed Challenges and Negative Sampling Strategies
- ☒ Planned Experiments

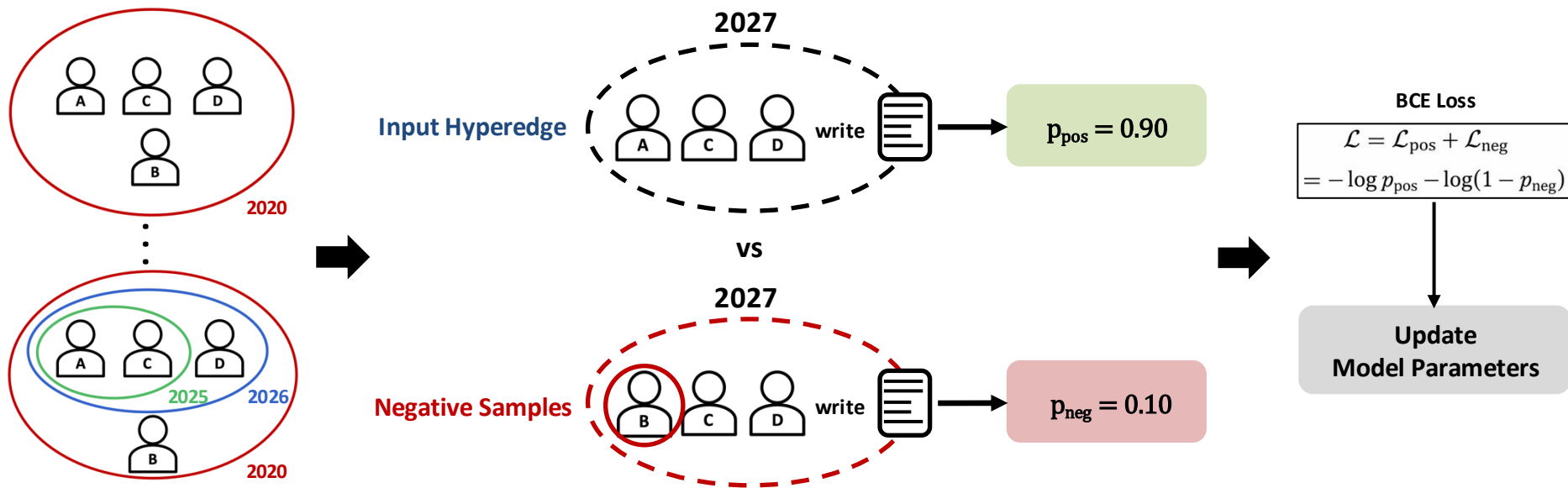
Overview of Dynamic Hypergraph Learning

- Learning time-evolving node and hyperedge representations
- Prediction of future hyperedge formation from historical interactions



Training Process in Dynamic Hypergraph Learning

- Node-replacement-based negative sampling
 - Replacing half of the nodes in a positive hyperedge with randomly sampled nodes
- Training and evaluation framework similar to dynamic graph learning
 - BCE-based binary classification, AUC and AP as evaluation metrics



Motivation for Harder Negative Sampling

- Near-perfect performance under standard random negative sampling

Transductive Setting.																
Approach	EE		NC		NS		UT		CB		TMS		CD		TSO	
	AUC	AP	AUC	AP	AUC	AP	AUC	AP	AUC	AP	AUC	AP	AUC	AP	AUC	AP
NHP HyperSAGCN CHESHIRE	63.17	66.87	82.39	80.72	81.38	82.17	82.33	81.44	80.27	77.82	79.20	78.53	-	-	-	-
	83.59	80.54	80.76	80.50	85.07	85.32	85.18	80.99	82.84	81.12	81.77	79.56	-	-	-	-
	82.27	81.39	84.91	82.24	86.30	83.18	82.75	81.96	86.81	83.66	-	-	-	-	-	-
CE-CAW	79.57	78.37	76.30	77.73	84.72	84.93	80.86	80.57	76.99	77.05	80.91	83.40	-	-	-	-
CE-Zebra	80.17	76.29	91.95	91.54	88.99	89.05	92.16	91.03	89.23	87.71	88.02	87.72	-	-	-	-
CE-Orca	78.62	78.35	87.92	87.39	86.54	86.77	83.25	80.13	87.97	87.72	72.61	67.31	-	-	-	-
CATWalk	80.47	82.87	98.72	98.71	90.64	91.96	93.51	94.98	88.15	88.66	90.63	91.26	-	-	-	-
FastHeP	89.93	92.82	99.84	99.01	94.57	93.52	97.44	96.54	98.69	98.51	97.89	96.61	91.38	90.52	93.89	93.78

- Potential variation in negative difficulty by replacement-node characteristics
- Focus on **temporal recency** and **interaction frequency**

Experiment Settings

☐ Models

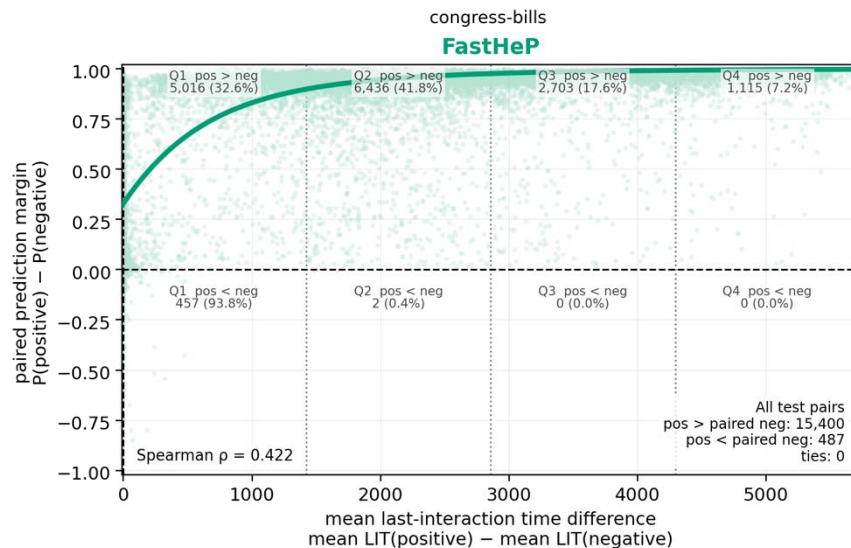
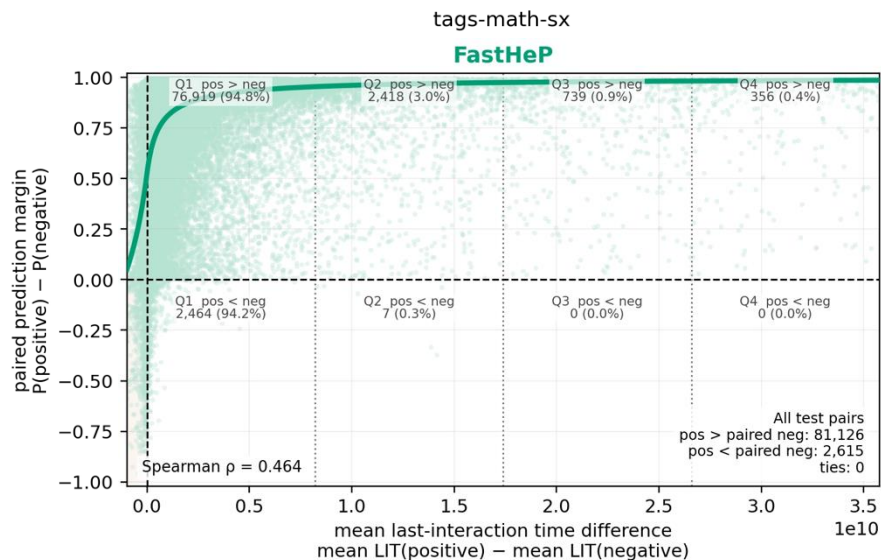
- Dynamic graph: JODIE, TGAT, TGN, GraphMixer
- Dynamic Hypergraph: FastHeP (KDD 2025)

☐ 6 datasets

- NDC-classes
- Contact-high-school
- Email-Enron
- Congress-bills
- Tags-math-sx
- Threads-ask-ubuntu

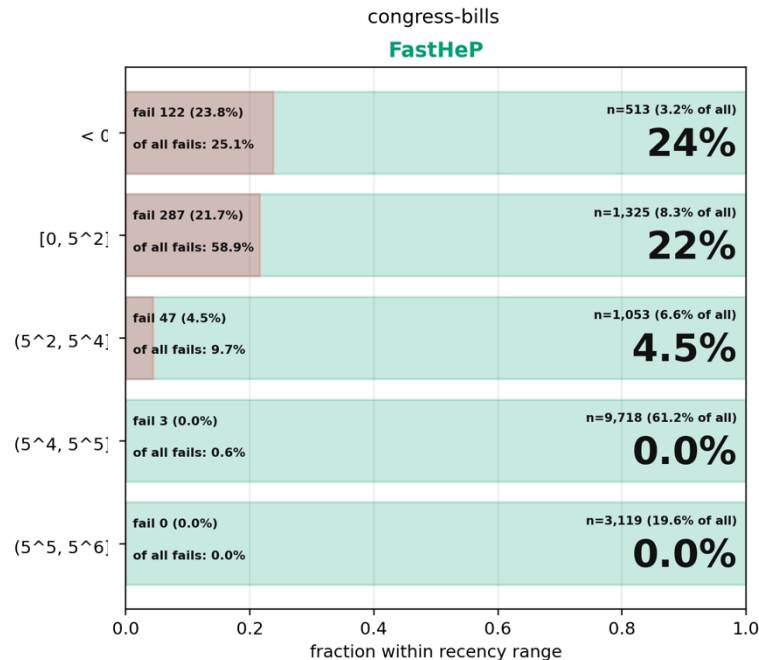
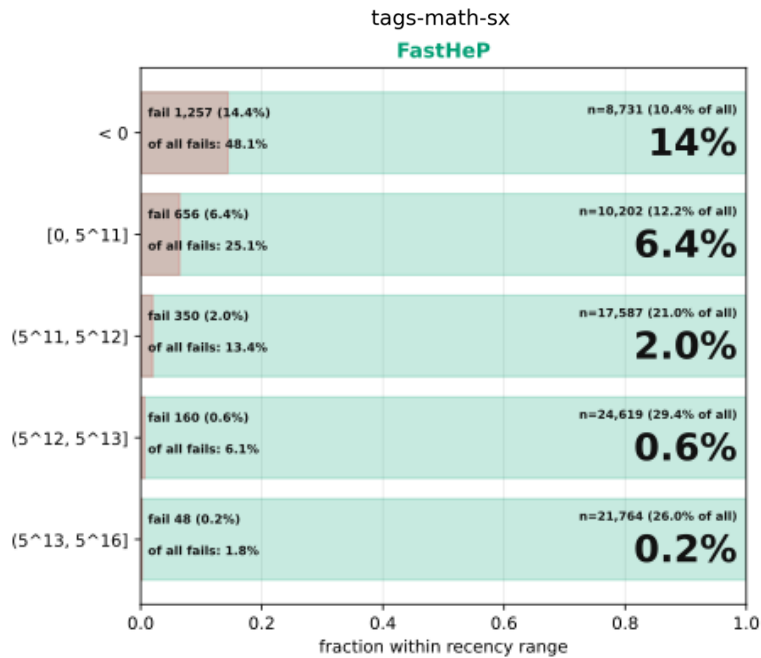
Observation 1 – Replacement-Node Recency

- Recency-based analysis of replacement nodes
- X-axis: Difference in mean latest interaction time
 - Difference between the mean latest interaction times of nodes in the positive and negative hyperedges
- Y-axis: Prediction score margin



Observation 1 – Ranking Failure by Node Recency

- Higher ranking-failure rates with recently active replacement nodes
- Stronger effect with a larger proportion of recent replacement nodes

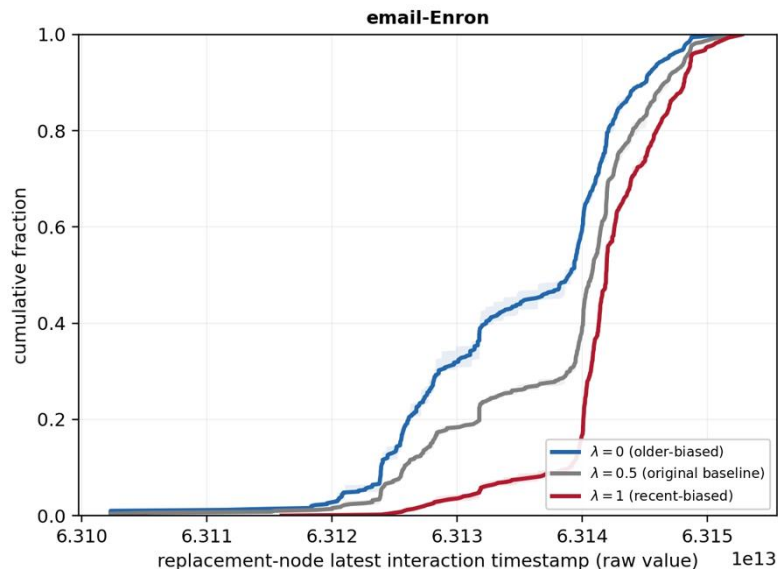


Recency-Aware Node Replacement

□ Replacement-node sampling based on latest interaction time

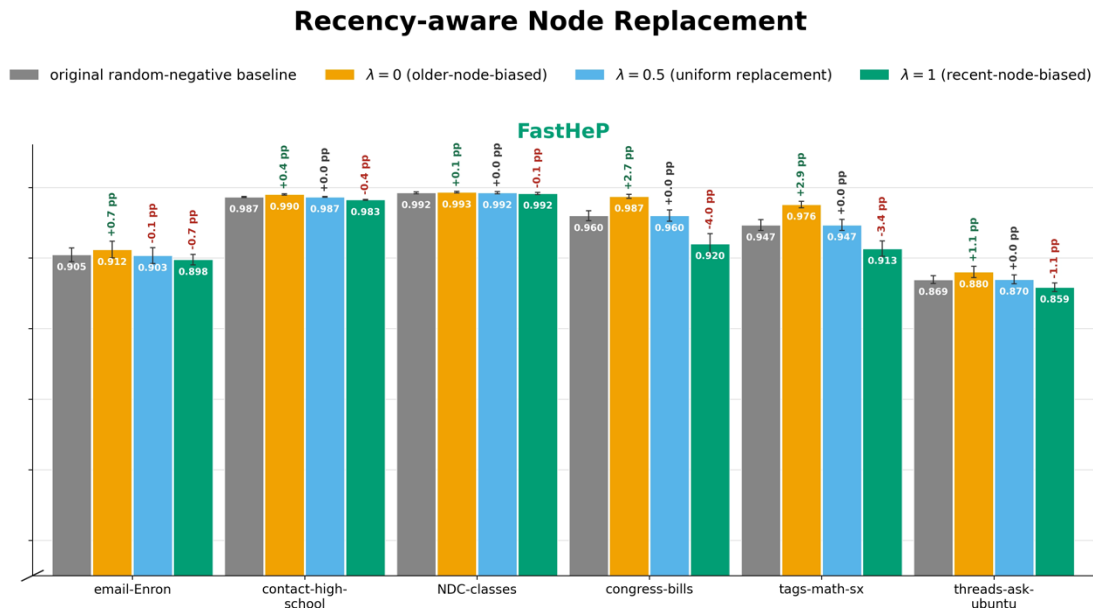
- Old-biased / uniform / recent-biased sampling
- Higher $\lambda \rightarrow$ stronger recent bias, $\lambda = 0.5 \rightarrow$ uniform

Replacement-node latest-interaction timestamp distributions (seen nodes)



Performance under Recency-Aware Node Replacement

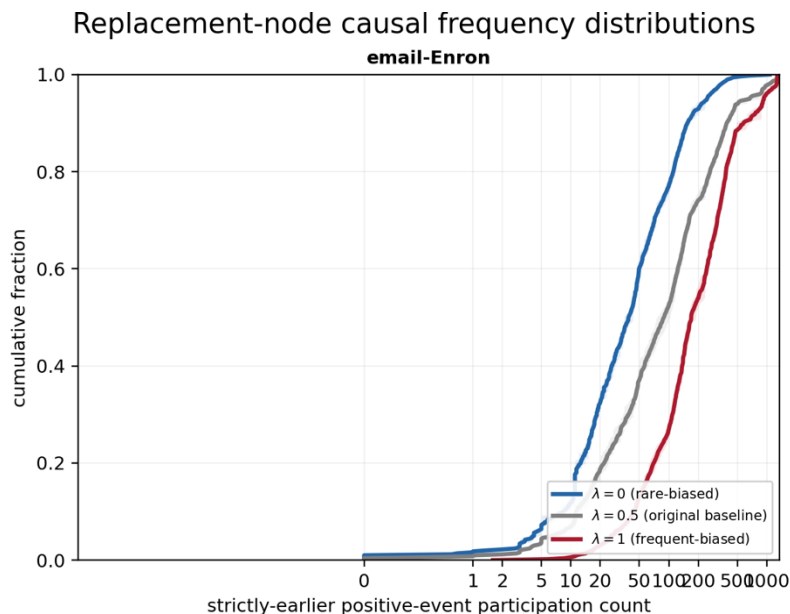
- Moderate performance degradation under recent-biased replacement
- Higher negative difficulty with recently active replacement nodes



Frequency-Aware Node Replacement

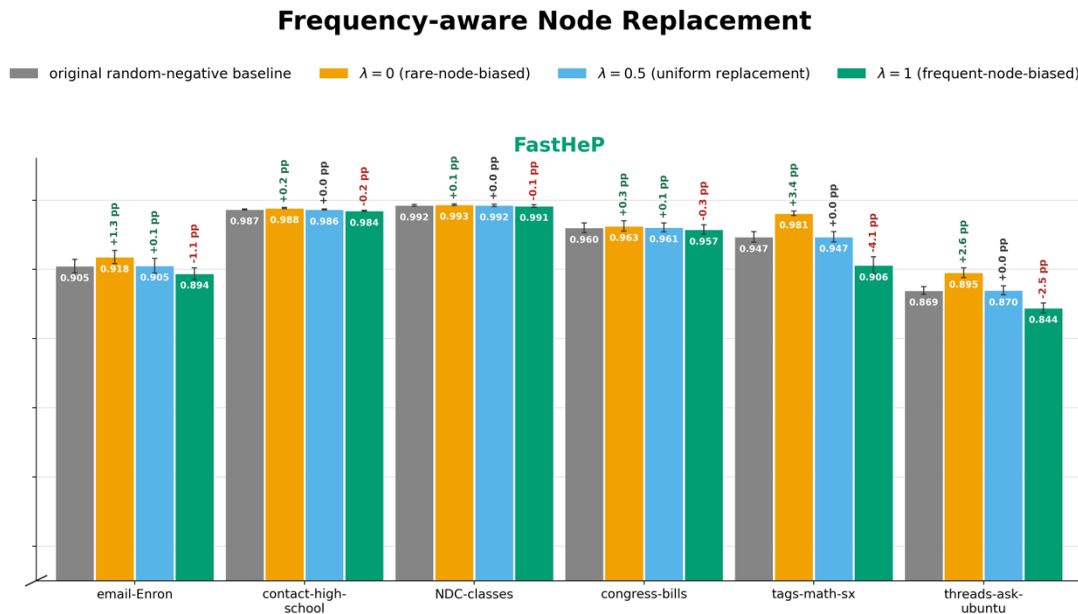
□ Replacement-node sampling based on interaction frequency

- Rare-biased / uniform / frequent-biased sampling
- Higher $\lambda \rightarrow$ stronger frequency bias, $\lambda = 0.5 \rightarrow$ uniform



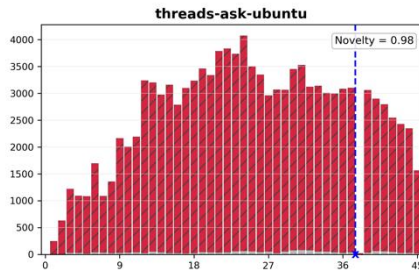
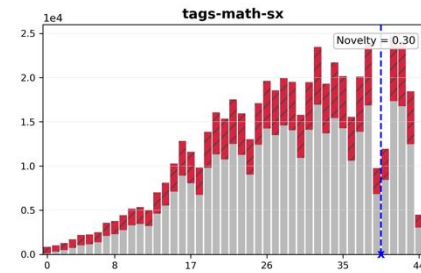
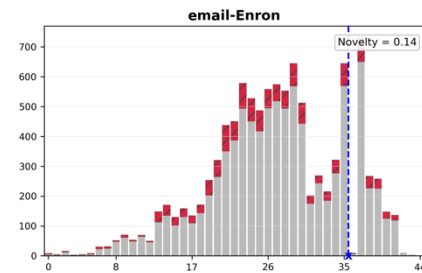
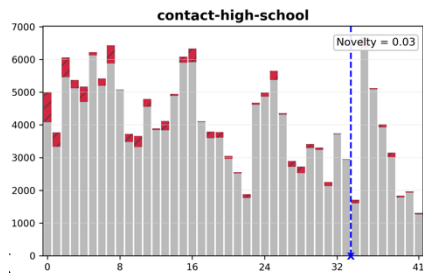
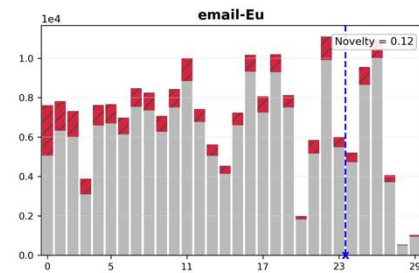
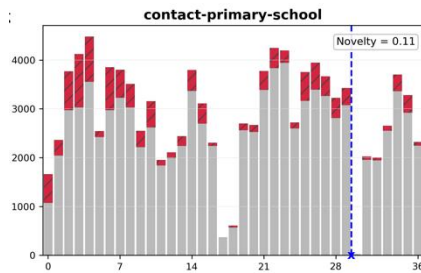
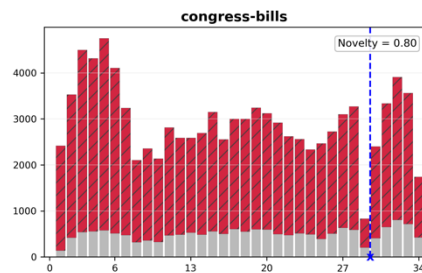
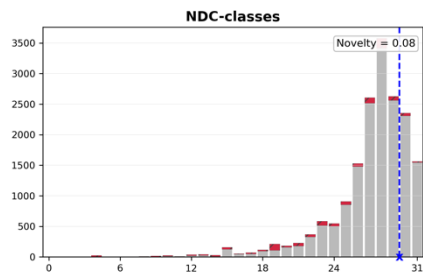
Performance under Frequency-Aware Node Replacement

- Moderate performance degradation under frequent-node-biased replacement
- Higher negative difficulty with frequently active replacement nodes



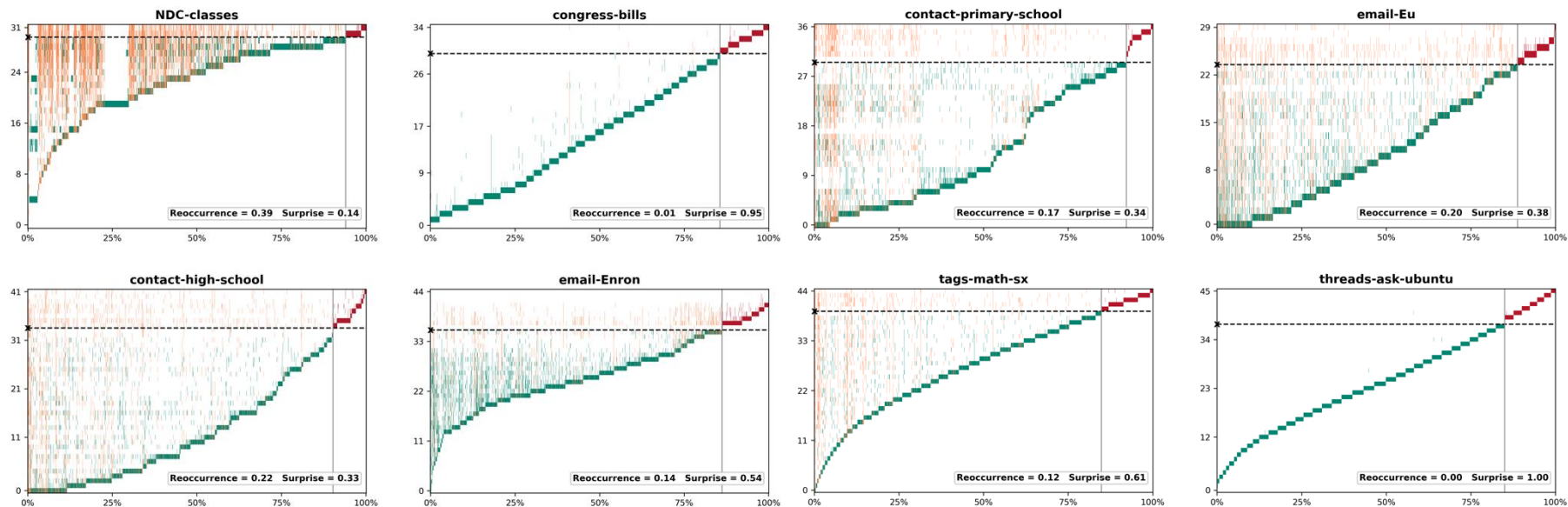
Observation 2 – Hyperedge Reoccurrence

- TEA-based analysis of repeated and newly observed hyperedges
- Substantial exact hyperedge reoccurrence across multiple datasets
 - Despite higher-order exact-match requirements



Observation 2 – Cross-Split Hyperedge Reoccurrence

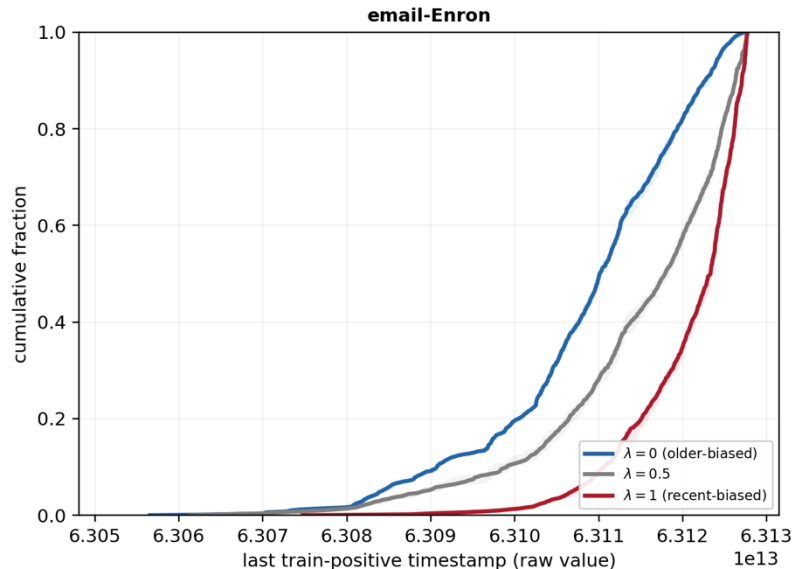
- TET-based analysis of train–test hyperedge reoccurrence
- Dataset-dependent reoccurrence and surprise patterns
- Substantial train–test overlap in several datasets



Recency-Aware Historical Negative Sampling

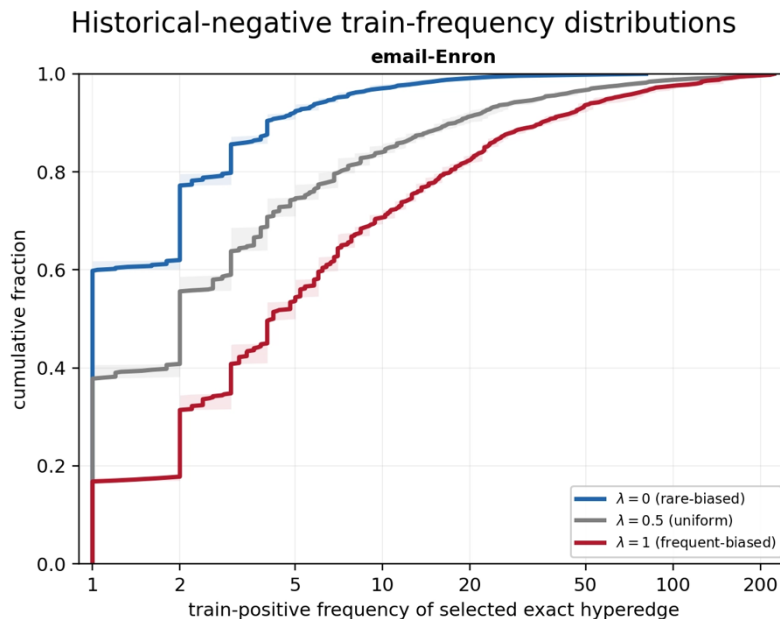
- Historical negatives from previously observed hyperedges
 - Old-biased / uniform / recent-biased sampling by last occurrence time
 - Higher $\lambda \rightarrow$ stronger recent bias, $\lambda = 0.5 \rightarrow$ uniform

Historical-negative last train-positive timestamp distributions



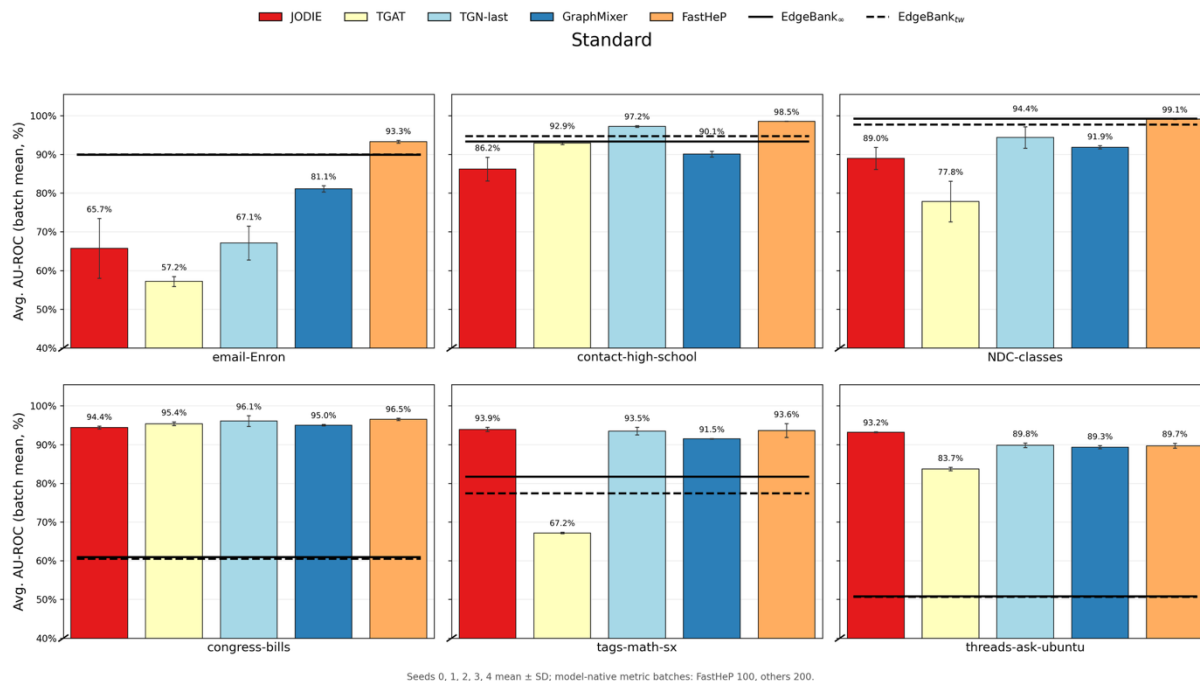
Frequency-Aware Historical Negative Sampling

- Historical negatives based on past hyperedge frequency
 - Rare-biased / uniform / frequent-biased sampling
 - Higher $\lambda \rightarrow$ stronger frequency bias, $\lambda = 0.5 \rightarrow$ uniform



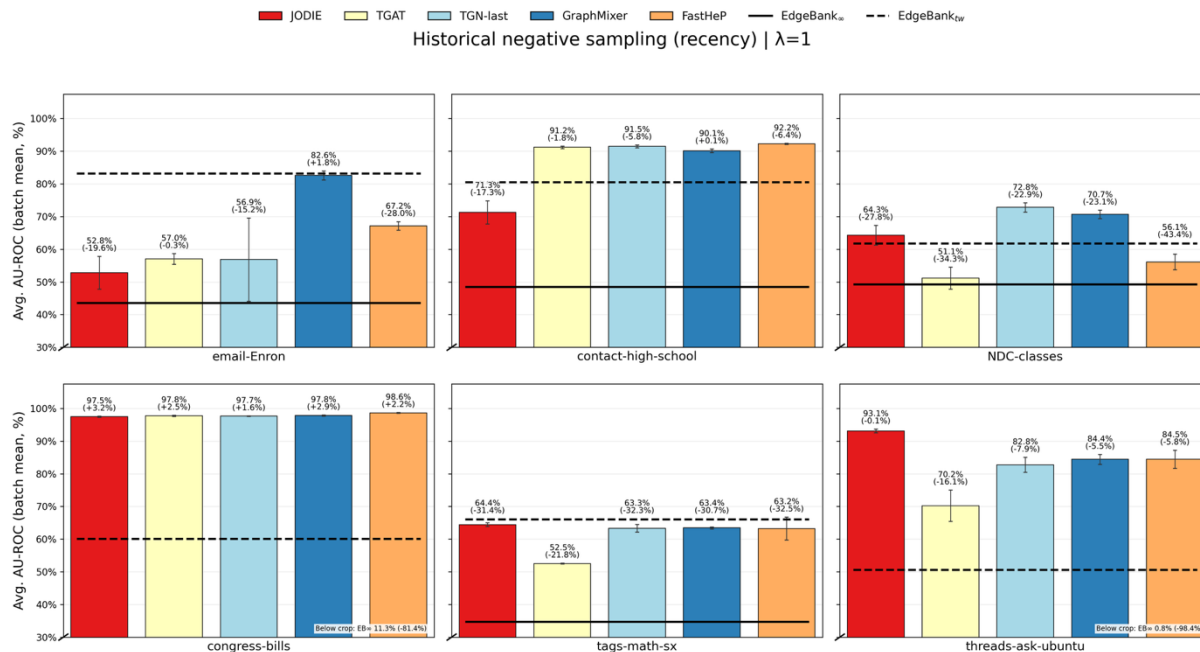
Performance under Standard NS

- Strong performance on several datasets under standard negative sampling
- Competitive performance of EdgeBank on highly recurrent datasets



Performance under Recency-Aware Historical NS

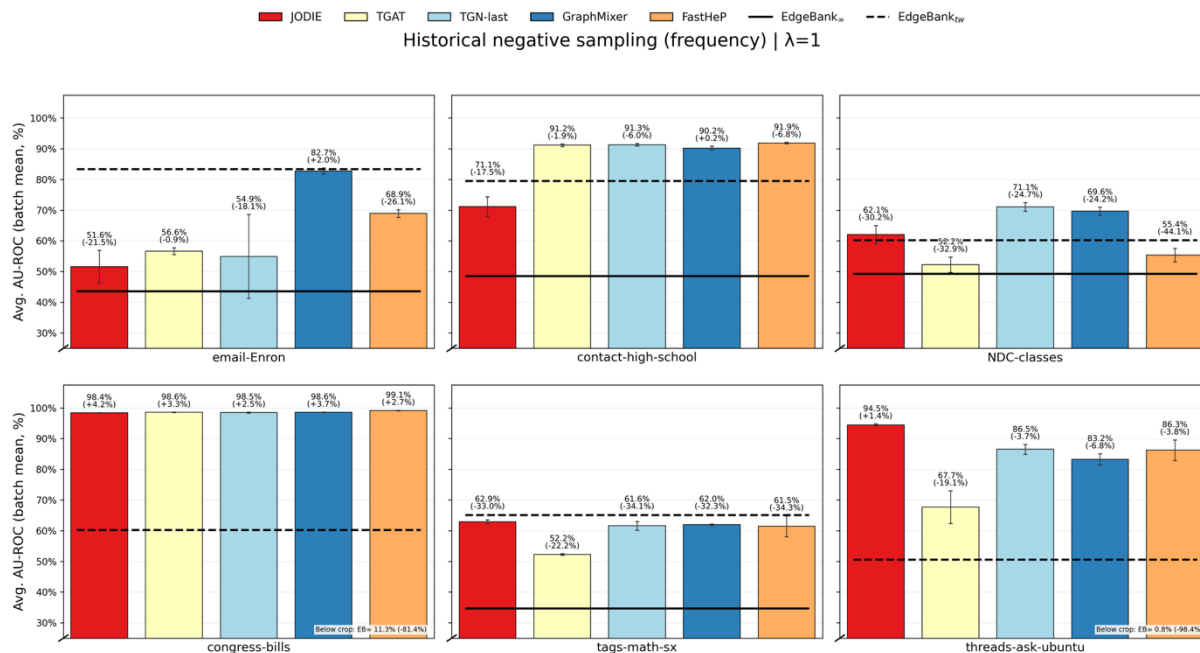
- Substantial performance degradation on 5 out of 6 datasets
- Congress-bills as an exception



Seeds 0, 1, 2, 3, 4 mean $\pm$ SD; model-native metric batches: FastHeP 100, others 200. Parentheses: relative change from standard.

Performance under Frequency-Aware Historical NS

- Substantial performance degradation on 5 out of 6 datasets
- Slightly less performance degradation overall than recency-aware sampling



Seeds 0, 1, 2, 3, 4 mean $\pm$ SD; model-native metric batches: FastHeP 100, others 200. Parentheses: relative change from standard.

Planned Experiments



- ☐ Confirmed difficulty of historical negative samples
- ☐ Training with historical hard negatives for improved robustness across evaluation settings
- ☐ Training with pre-generated offline historical negatives
- ☐ Extension to online, history-aware negative sampling during training

Thank you

Table 12: Statistics of negative edges used during the test phase in historical or inductive NS setting.

Dataset	Historical NS		Inductive NS		# Total Negative Edges
	# Random	# Historical	# Random	# Inductive	
Wikipedia	0	23,621	1,018	22,603	23,621
Reddit	0	100,867	0	100,867	100,867
MOOC	0	61,763	351	61,412	61,768
LastFM	0	193,966	0	193,966	193,966
Enron	0	18,785	3,689	15,096	18,785
Social Evo.	0	314,924	268,958	45,966	314,924
UCI	0	8,976	402	8,574	8,976
Flights	0	287,824	18,800	269,024	287,824
Can. Parl.	0	10,113	7200	2,913	10,113
US Legis.	0	9,804	5,000	4,804	9,804
UN Trade	0	61,595	20,800	40,795	61,595
UN Vote	0	155,119	73,167	81,952	155,119
Contact	0	363,780	5,227	358,553	363,780

Singleton-free train-event calendar / elapsed-time distributions

